ALESSANDRO AGAPITO

(Astro) physicist

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Astrophysicist with a background in theory, statistics, and computational physics. **Research experience** includes: gravitational waves (GWs), cosmology, tests of General Relativity (GR), Bayesian inference, machine learning.

EDUCATION

Ph.D. in Gravitational-Wave Cosmology

Aix-Marseille Université, Centre de Physique Théorique

Period: oct 2024 - oct 2027 — Marseille, France

- **Thesis project**: "Gravitational-wave cosmology between current and future-generation detectors". Development of advanced Bayesian methods to test cosmology and fundamental physics from gravitational waves, with applications to current and future observations. **Supervisor**: Michele Mancarella
- **Responsibilities**: Coordinator of the Cosmology group's weekly meetings

M.Sc. in Astronomy and Astrophysics

Sapienza University of Rome

Period: sep 2023 - sep 2025 — Rome, Italy

- **Final mark**: 110/110 with honors (weighted averaged mark 29.7/30.0)
- **Thesis project**: "Gravitational waves inspiral tests of General Relativity in the framework of the Einstein Telescope". Main results included in 1 publication (A. Abac et. al, 2026, "The Science of the Einstein Telescope") and presented at 1 international conference and 1 national meeting. **Supervisors**: Paolo Pani, Andrea Maselli

B.Sc. in Physics

Sapienza University of Rome

Period: sep 2019 - oct 2022 — Rome, Italy

- **Final mark**: 110/110 (weighted averaged mark 28.0/30.0)
- **Thesis project**: "Stelle degeneri". **Supervisor**: Alfredo Leonardo Urbano

INTERNATIONAL AND NATIONAL COLLABORATIONS

- **The Einstein Telescope** (ET) collaboration, Observational Science Board (OSB)
- **Virgo** collaboration within LIGO-Virgo-KAGRA (LVK), Compact Binary Coalescence (CBC) group
- **Groupement de Recherche Ondes Gravitationnelles** (GdR), national working group interested in the exploration of the Universe with GWs

PUBLICATIONS AND RESEARCH OUTPUT

Selected publications (iNSPIRE, ORCID)

- **A. Agapito**, M. Mancarella, "Gravitational-wave dark sirens as astrophysical probes: inferring galaxy–merger relations and identifying individual hosts", arXiv:2609.38110 (submitted to PRL), **leading author**
- **A. Agapito**, V. De Renzis, M. Mancarella, "Gravitational-wave constraints on H_0 are robust to putative redshift evolution in the binary black hole mass spectrum at current sensitivity", Phys.Rev.D 114 (2026) 6, 063503, **leading author**
- V. De Renzis, **A. Agapito**, M. Mancarella, "A weakly modelled view of the joint compact-binary mass plane: population structure and spectral-siren cosmology", arXiv:2610.10535(submitted to PRD), **corresponding author**
- M. Mancarella, V. De Renzis, **A. Agapito**, "A gaussian–process reconstruction of gravitational–wave friction from dark standard sirens", (ready for submission)
- M. Tagliazucchi, M. Moresco, **A. Agapito**, M. Mancarella, S. Ferraiuolo, S. Mastrogiovanni, N. Borghi, F. Pannarale, D. Bonacorsi, "Pushing spectral siren cosmology into the third-generation era: a blinded mock data challenge", Astron.Astrophys. 711 (2026) A120

Wide group projects and collaboration papers

- **1** ET wide group project ($\sim$ 30 authors) + **1** ET collaboration paper (both with active contribution)
- **40** LVK collaboration papers (of which **1** with active contribution as a member of the analysts team)

Selected talks and presentations

- **4 international conferences** (2026 LVK Collaboration Meeting, 2026 PONT, 2026 and 2025 ET Symposium)
- **2 national meetings** (2025 Journée du groupe de travail sur la cosmologie du GdR ondes gravitationnelles, 2024 Huitième Assemblée Générale du GdR Ondes Gravitationnelles)

Schools and workshops

- "AIPHY2: Bayesian Statistics School", 2026
- "Fourth MaNiTou Summer School on Gravitational Waves: A new window to the Universe", 2025
- "Paris workshop on Bayesian Deep Learning for Cosmology and Time Domain Astrophysics", 2025
- "Gravitational Wave Open Data Workshop #7", 2024
- "JENAS Initiative: Gravitational Wave Probes of Fundamental Physics", 2024

GRANTS AND SCOLARSHIPS

- **Ph.D. scholarship** "Cosmology and Astrophysics with Gravitational Waves and Galaxy Catalogs", A*Midex foundation - Initiative of excellence grant, N. AMX-22-CEI-02 COS-MOAGG. Advisor and Principal Investigator: Michele Mancarella
- **Excellence Programme** at Sapienza University: Advanced courses programme, for 5 selected students (2024)
- **3 tutoring grants** and **2 collaboration grants** at Sapienza University of Rome (between 2022 and 2024)

UNDERGRADUATE TEACHING AND MENTORING

Undergraduate student's mentoring

Aix-Marseille Université, Centre de Physique Théorique

Period: October-December 2026 — Marseille, France

- M. Berva, University of Milano Bicocca/ Aix-Marseille Université, **Master's Thesis**

Mécanique (JPE402A)

Polytech

Period: May 2026 — Marseille, France

- **Mechanics of solids and liquids**. Assistance and exercises during lab practices in French (30h, $\sim$ 60 participants).

Gravitational Wave Open Data Workshop #8

Aix-Marseille Université

Period: May 2025 — Marseille, France

- GW data analysis. Assistance and exercises during lessons (12h, $\sim$ 10 participants)

Tutoring grants

Faculty of Mathematics and Natural Sciences

Period: November 2023 - June 2024 — Rome, Italy

- **Mechanics Laboratory** (Bachelor's degree, Physics). Lessons and laboratories (75h, $\sim$ 50 participants)
- **Mathematics Institutions I** (Bachelor's degree, Chemistry). Lessons (75h, $\sim$ 70 participants)
- **Physics and Statistics** (Bachelor's degree, Engineering). Lessons (30h, $\sim$ 20 participants)

SKILLS AND LANGUAGES

Computing and Libraries

- **Python** (> 1000 lines), **C** (> 500 lines), **Mathematica** (> 100 lines), **Bash**, **Git**, **OpenMP**, **MPI**
- **NumPy**, **SciPy**, **Matplotlib**, **JAX**, **pytensor**, **pytorch**, **PyMC**, **NumPyro**

Languages

- **Italian** (Native)
- **English** (Fluent, B2 level within FCE in 2018 and English language exam at Sapienza University of Rome in 2023)
- **French** and **Spanish** (Intermediate, B1 level)